

Phase 4 — Triage Workflow

1. Triage Playbook

1.1 Purpose

The purpose of this triage playbook is to provide a consistent and structured process for reviewing security alerts generated by Wazuh. The playbook helps the analyst determine the severity of an alert, investigate the activity, decide whether the alert should be escalated or closed, and document the reasoning behind the decision.

The goal is to ensure that alerts are handled quickly and consistently while reducing unnecessary escalation of legitimate or authorized activity.

1.2 Severity Classification

Severity is assigned according to the **Wazuh rule level** configured for the detection.

Severity	Wazuh Level	Description
Low	1–4	Informational or low-risk activity with limited security impact.
Medium	5–7	Suspicious activity that requires investigation but presents a moderate level of risk.
High	8–11	Significant suspicious activity that may indicate persistence, unauthorized execution, credential access, or other security risks.
Critical	12–15	Severe activity that may indicate credential theft, active attacks, unauthorized access, or potential system compromise.

The Wazuh rule level provides the initial severity classification. The analyst should still review the alert context and supporting evidence before making a final triage decision.

Examples from the Practice Batch

- **Level 8 — Cron Job Creation:** High
- **Level 9 — Systemd Service Modification:** High
- **Level 10 — Command Execution:** High
- **Level 10 — Shadow File Access:** High
- **Level 10 — Suspicious Execution:** High
- **Level 12 — Credential Dumping:** Critical
- **Level 12 — SSH Brute Force:** Critical

1.3 Escalation Criteria

An alert should be **escalated** to a higher-level analyst or incident response team when one or more of the following conditions are identified:

- Evidence of successful unauthorized access or system compromise.
- Evidence of credential theft or credential dumping.
- Multiple related alerts indicate a possible attack chain.
- Suspicious persistence mechanisms are identified and cannot be validated as legitimate.
- A brute-force attack is followed by a successful authentication.
- Malicious activity is confirmed or cannot reasonably be ruled out.
- The activity affects a privileged account, critical system, or sensitive information.
- The activity is ongoing or appears to be causing active impact.
- The Tier 1 analyst does not have sufficient information or access to safely determine the disposition.

Closure Criteria

An alert may be **closed** when:

- The activity is confirmed to be legitimate and authorized.
- The activity was intentionally generated as part of an approved security test or lab simulation.
- Investigation finds no evidence of actual compromise.
- The detection was triggered by benign activity.
- The alert has been investigated and no further action is required.
- The alert is a duplicate of an already investigated event and contains no new information.

The analyst must document the reason for closing the alert.

1.4 Triage Workflow

The following workflow should be followed for each alert:

Step 1 — Review the Alert

Review the alert information in Wazuh, including:

- Alert ID
- Timestamp
- Rule ID
- Rule description
- Wazuh rule level
- Affected host or agent
- User involved
- Log source
- MITRE ATT&CK technique, if available

Step 2 — Validate the Alert

Confirm that the detected activity actually occurred by reviewing the original log or event.

The analyst should determine whether:

- The rule fired correctly.
- The activity was generated by a legitimate process or user.
- Similar events occurred around the same time.

Step 3 — Investigate the Activity

Gather additional context relevant to the alert.

Depending on the detection, the analyst may investigate:

- User involved
- Process or command executed
- File or service affected
- Source IP address
- Authentication activity
- Related events
- Activity before and after the alert
- Whether the activity was authorized

Step 4 — Determine the Detection Outcome

The analyst determines whether the detection accurately identified the activity.

Possible outcomes include:

- **True Positive:** The detection correctly identified the activity described by the rule.
- **False Positive:** The rule triggered on activity that was not relevant to the intended security detection.
- **Insufficient Evidence:** There is not enough information to confidently determine the nature of the activity.

Step 5 — Assign Severity

Assign the alert a severity of:

Low, Medium, High, or Critical

using the severity classification defined in this playbook.

Step 6 — Determine Disposition

The analyst decides whether the alert should be:

- **Closed:** No further action is required.
- **Escalated:** Additional investigation or response is required.

Step 7 — Document the Decision

The analyst records the investigation findings, severity, detection outcome, disposition, escalation decision, and reasoning.

1.5 Documentation Requirements

Each triaged alert should contain the following information:

- Alert ID
- Timestamp
- Rule ID
- Rule description
- Affected host/agent
- Wazuh rule level
- MITRE ATT&CK technique, where applicable
- Investigation findings
- Triage severity
- Detection outcome
- Disposition
- Escalation decision
- Reasoning

The reasoning should clearly explain **what was observed, what was determined during investigation, and why the alert was closed or escalated.**

2. Completed Triage Log

Practice Batch Context:

The practice batch contains both intentionally simulated security activities and system-generated benign events. The alerts were reviewed using the triage workflow defined in this playbook. The purpose was to determine whether Wazuh correctly detected the activity and whether the alert required escalation or closure.

Alert 1 — Command Execution

Alert ID: 1787687800.203443

Timestamp: Aug 25, 2026 @ 20:56:40

Rule ID: 100407

Wazuh Level: 10

MITRE ATT&CK: T1059.004 — Unix Shell

Log Source: /var/log/audit/audit.log

Detection: Interactive Bash execution on terminal (pts0) by /usr/bin/bash

Investigation

Wazuh detected the execution of an interactive Bash shell. The activity was intentionally generated as part of the authorized lab simulation. No evidence of unauthorized access, malicious commands, or system compromise was identified from the available alert information.

Triage Decision

- **Severity:** High
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected the interactive Bash execution. The activity was intentionally generated as part of an authorized lab simulation, and no evidence of actual compromise was identified. The alert was therefore closed without escalation.

Alert 2 — Credential Dumping

Alert ID: 1787674059.21184

Timestamp: Aug 25, 2026 @ 17:07:39

Rule ID: 100409

Wazuh Level: 12

MITRE ATT&CK: T1003.008 — OS Credential Dumping

Log Source: /var/log/audit/audit.log

Detection: Possible credential dumping through /proc memory access or shadow file inspection

Investigation

Wazuh detected activity associated with possible credential dumping. The activity was intentionally generated as part of the authorized lab simulation. The alert does not provide evidence of actual credential theft, unauthorized access, or system compromise.

Triage Decision

- **Severity:** Critical
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected activity associated with possible credential dumping. Because the activity was intentionally generated as part of an authorized lab simulation and no evidence of actual credential theft or compromise was identified, the alert was closed without escalation.

Alert 3 — Systemd Service Modification

Alert ID: 1787699082.376709

Timestamp: Aug 26, 2026 @ 00:04:42

Rule ID: 100405

Wazuh Level: 9

MITRE ATT&CK: T1543.002 — Systemd Service

Log Source: Syscheck

Detection: /etc/systemd/system/atomic_test.service added

Investigation

Wazuh detected the creation of a systemd service file. Systemd services can be used legitimately but can also be abused as a persistence mechanism. The activity was intentionally generated as part of the authorized lab simulation.

Triage Decision

- **Severity:** High
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected the creation of a systemd service file, which can be used as a persistence mechanism. The activity was intentionally generated as part of an authorized lab simulation, and no evidence of actual malicious persistence or compromise was identified. The alert was therefore closed without escalation.

Alert 4 — Cron Job Creation

Alert ID: 1787698835.371033

Timestamp: Aug 26, 2026 @ 00:00:36

Rule ID: 100402

Wazuh Level: 8

MITRE ATT&CK: T1053.003 — Cron

Log Source: Syscheck

Detection: /etc/cron.d/attacksim_final added

Investigation

Wazuh detected the creation of a cron job. Cron can be used legitimately for scheduled tasks but can also be abused to establish persistence. The activity was intentionally generated as part of the authorized lab simulation.

Triage Decision

- **Severity:** High
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected the creation of a cron job, a mechanism that can be abused for persistence. The activity was intentionally generated as part of an authorized lab simulation, and no evidence of actual malicious persistence or system compromise was identified. The alert was therefore closed without escalation.

Alert 5 — Shadow File Access Attempt

Alert ID: 1787674365.25309

Timestamp: Aug 25, 2026 @ 17:12:45

Rule ID: 100410

Wazuh Level: 10

MITRE ATT&CK: T1552.001 — Credentials from Password Stores

Log Source: /var/log/audit/audit.log

Detection: Unauthorized access attempt to /etc/shadow by /usr/lib/cargo/bin/coreutils/cat

Investigation

Wazuh detected an attempt to access the sensitive /etc/shadow file. The activity was intentionally generated as part of the authorized lab simulation. No evidence of successful credential theft or system compromise was identified.

Triage Decision

- **Severity:** High
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected an attempt to access the sensitive `/etc/shadow` file. The activity was intentionally generated as part of an authorized lab simulation, and no evidence of successful credential theft or system compromise was identified. The alert was therefore closed without escalation.

Alert 6 — Possible SSH Brute Force Attack

Timestamp: Jul 30, 2026

Rule ID: 100300

Wazuh Level: 12

Rule Frequency: 5

Rule Fired: 1

Rule Description: Possible SSH Brute Force Attack — 5+ failed logins from the same IP within 5 minutes

Host: osboxes

Source IP: 192.168.35.132

Target User: fakeuser

Log Source: SSH/sshd

Investigation

Wazuh detected repeated failed SSH authentication attempts from 192.168.35.132 targeting the invalid user `fakeuser`. The rule was designed to identify five or more failed login attempts from the same IP within five minutes.

The activity was intentionally generated as part of the authorized lab simulation. The provided logs show failed authentication attempts and do not show a successful login or evidence of compromise.

Triage Decision

- **Severity:** Critical
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected repeated failed SSH authentication attempts matching the configured brute-force threshold. The activity was intentionally generated as part of an

authorized lab simulation, and the provided logs show no successful authentication or evidence of compromise. The alert was therefore closed without escalation.

Alert 7 — Suspicious Execution

Timestamp: Jul 30, 2026 @ 03:18:14.933

Rule ID: 100003

Wazuh Level: 10

Rule Fired: 2

Host: osboxes

Agent: hayzed

Agent ID: 001

Agent IP: 192.168.35.133

User: osboxes

Destination User: root

TTY: tty1

Working Directory: /home/osboxes

Command: /tmp/test_script

Detection: Binary or script executed from /tmp/

Investigation

Wazuh detected the execution of /tmp/test_script using sudo, with the destination user being root. Execution from a temporary directory can be suspicious because attackers may use temporary directories to store and execute malicious files.

The activity was intentionally generated as part of the authorized lab simulation.

Triage Decision

- **Severity:** High
- **Detection Outcome:** True Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

Wazuh correctly detected execution of /tmp/test_script with root privileges. Execution from a temporary directory can indicate suspicious activity and therefore warranted investigation. The activity was intentionally generated as part of an authorized lab simulation, and no evidence of actual compromise was identified. The alert was therefore closed without escalation.

Alert 8 — Interface Promiscuous Mode

Alert ID: 1787510303.187600

Timestamp: Aug 23, 2026 @ 19:31:11

Host/Location: 10.0.3.1 (pfSense)

Decoder: kernel

Input Type: log

Event: pflog0: promiscuous mode enabled

Wazuh Level: 5

Investigation

Wazuh detected an event indicating that the pflog0 interface entered promiscuous mode. Promiscuous mode can be associated with network monitoring and packet inspection activities. In this case, the event was generated by the pfSense system as part of normal network monitoring activity. No evidence of malicious packet capture, unauthorized access, or system compromise was identified.

Triage Decision

- **Severity:** Medium
- **Detection Outcome:** False Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

The event was generated by legitimate network monitoring activity on the pfSense system. Although promiscuous mode can be associated with suspicious network activity, no evidence of malicious behavior was identified. The alert was therefore classified as a false positive and closed without escalation.

Alert 9 — AppArmor Access Denied

Alert ID: 1787481660.20984

Timestamp: Aug 23, 2026 @ 11:41:00

Host: hayzed

Log Source: journald

Decoder: kernel

Status: DENIED

Operation: open

Profile: who

File: /usr/share/coreutils/locales/uucore/en-US.ftl

Process: who

PID: 2822

Wazuh Level: 5

Investigation

Wazuh detected an AppArmor access denial involving the legitimate who utility. AppArmor blocked the process from accessing the specified file according to the system's security policy. The event represents normal security-policy enforcement rather than successful malicious activity.

Triage Decision

- **Severity:** Medium
- **Detection Outcome:** False Positive
- **Disposition:** Closed
- **Escalation:** No

Reasoning

The event was generated because AppArmor blocked a file-access request made by the legitimate who utility. The security control successfully prevented the access, and there was no evidence of malicious activity or system compromise. The alert was therefore classified as a false positive and closed without escalation.

3. Final Triage Summary

#	Alert	Rule ID	Wazuh Level	Severity	Outcome	Disposition	Escalate
1	Command Execution	100407	10	High	True Positive	Closed	No
2	Credential Dumping	100409	12	Critical	True Positive	Closed	No
3	Systemd Service Modification	100405	9	High	True Positive	Closed	No
4	Cron Job Creation	100402	8	High	True Positive	Closed	No
5	Shadow File Access	100410	10	High	True Positive	Closed	No
6	SSH Brute Force	100300	12	Critical	True Positive	Closed	No
7	Suspicious Execution	100003	10	High	True Positive	Closed	No

#	Alert	Rule ID	Wazuh Level	Severity	Outcome	Disposition	Escalate
8	Interface Promiscuous Mode	5104	5	Medium	False Positive	Closed	No
9	AppArmor Access Denied	52002	5	Medium	False Positive	Closed	No